

Unified Symphonix-Health Agentic Architecture

This revision replaces the earlier repo-inference based report with an architecture brief grounded in the local repositories under `C:\Users\hgeec\github`. It keeps the original layered intent, but corrects ownership, data flow, agent routing, clinical safety, and evidence boundaries.

10

Core platform repos treated as explicit architecture inputs.

0 PHI

GHARRA remains a registry and trust plane, not a patient-data store.

13

Nexus route-admission checks before transport opens.

160+

BulletTrain services form the healthcare exchange fabric.

Primary correction. The previous PDF said several rows were based on "name inference" because repo access failed. This version treats local repo files as available evidence. Where a repository exposes an actual specification, requirement, route, test, or stated boundary, this report uses that instead of inferred responsibility.

Runtime rule

Agents augment service APIs. They do not replace REST, FHIR, X12, HL7v2, DICOM, or event-bus contracts.

Data rule

Services own domain data. BulletTrain owns cross-system exchange, audit, provenance, transformation, consent enforcement, and AI grounding.

Clinical rule

Clinical agents pass through CSAA classification, GHARRA registration, and Nexus A2A coordination before production use.

What changed from the original PDF

1. BulletTrain is the exchange plane, not a generic base layer

BulletTrain is the HIE and orchestration backbone for clinical and operational exchange. It owns FHIR/EventBus/API-gateway mediated movement, identity, consent, audit, provenance, routing, validation, and AI grounding across sibling systems.

Evidence: BulletTrain/README.md:33,43,51,60,65,67-69,784-785; symphonix-health-docs/strategy/bullettrain-integration-doctrine.md:11,56,100,117-119,161.

2. GHARRA is not an execution bus

GHARRA is the federated zero-trust registry and routing authority. It stores agent metadata, trust material, route policy, and A2A agent cards. It must not store patient data.

Evidence: global-agent-registry/README.md:26,91-101,139-166,209,259; global-agent-registry/CLAUDE.md:11,143-145,229,259-264.

3. Nexus validates and transports admitted calls

Nexus A2A provides JSON-RPC 2.0 agent transport and route-admission enforcement. The repo explicitly states Nexus does not perform discovery, store registry records, or decide which agent to call.

Evidence: nexus-a2a-protocol/README.md:26,71-73,144-166,189-190; nexus-a2a-protocol/docs/gharra_integration.md:14,18-21,228-231.

4. CSAA is a production gate for clinical agents

CSAA classifies clinical scope and builds safety cases, hazard logs, release decisions, and residual-risk checks. It is a runtime governance dependency for clinical-agent release, not an optional appendix.

Evidence: csaa/csaa/scope.py:1-16,142-217; csaa/csaa/case_builder.py:55-117,160-199; csaa/demo/pacs_ris_image_ai/safety/RELEASE_DECISION.md:12-64.

Agent and evidence boundaries

5. SignalBox is test, evidence, and attestation tooling

SignalBox exposes browser control, structured assertions, persona switching, visual regression, scenario execution, and GHARRA-anchored clinical-signal capture. It should appear in the evidence and control plane, not as a generic agent.

Evidence: signalbox-mcp/versions/v2/README.md:8-20,41,103-145,181-227,278-308; signalbox-mcp/INTEROP.md:9-15,101-110.

6. CAID and health-agent-workspace are not patient runtime services

CAID is the SDLC, requirements, architecture, matrix, and multi-agent engineering system. Health Agent Workspace provides market, pilot, and research grounding. They inform build priority and evidence quality; they are not clinical data stores.

Evidence: caid-agent/README.md:31-56,92-127,235-247,725-728; health-agent-workspace/GHARRA_Healthcare_AI_Agent_Market_Intelligence.md:8-10,360-381.

Architectural consequence. The platform should be drawn as a set of governed control planes, not as one agent brain. BulletTrain, GHARRA, Nexus, CSAA, SignalBox, Bridge SDK, Prompt Engine, CAID, and Health Agent Workspace each carry a separate accountability boundary.

Evidence-backed platform map

Clinical and operational value: safer workflows, governed automation, measurable outcomes

Applications and workflow action

Provider portal, citizen portal, triage, scheduling, LIS, PACS/RIS, pharmacy, claims, ERP, HMIS, specialty systems



Agent execution and evidence control

Prompt Engine, Symphonix Bridge SDK, SignalBox MCP, CAID-generated requirements and matrices, human review queues

Agent loop: context -> plan -> tool call -> review -> audit -> outcome measurement



Agent trust, discovery, safety, and transport

CSAA classifies clinical scope. GHARRA resolves signed capability cards and trust bundles. Nexus admits and transports JSON-RPC calls.

Nexus validates records. GHARRA owns registry. BulletTrain and SignalBox orchestrate.



Interoperability, identity, consent, and audit fabric

BulletTrain API Gateway, FHIR EventBus, Shared Health Record, terminology, audit store, observability, consent governance

All sibling-to-sibling platform exchange flows through BulletTrain-mediated contracts.



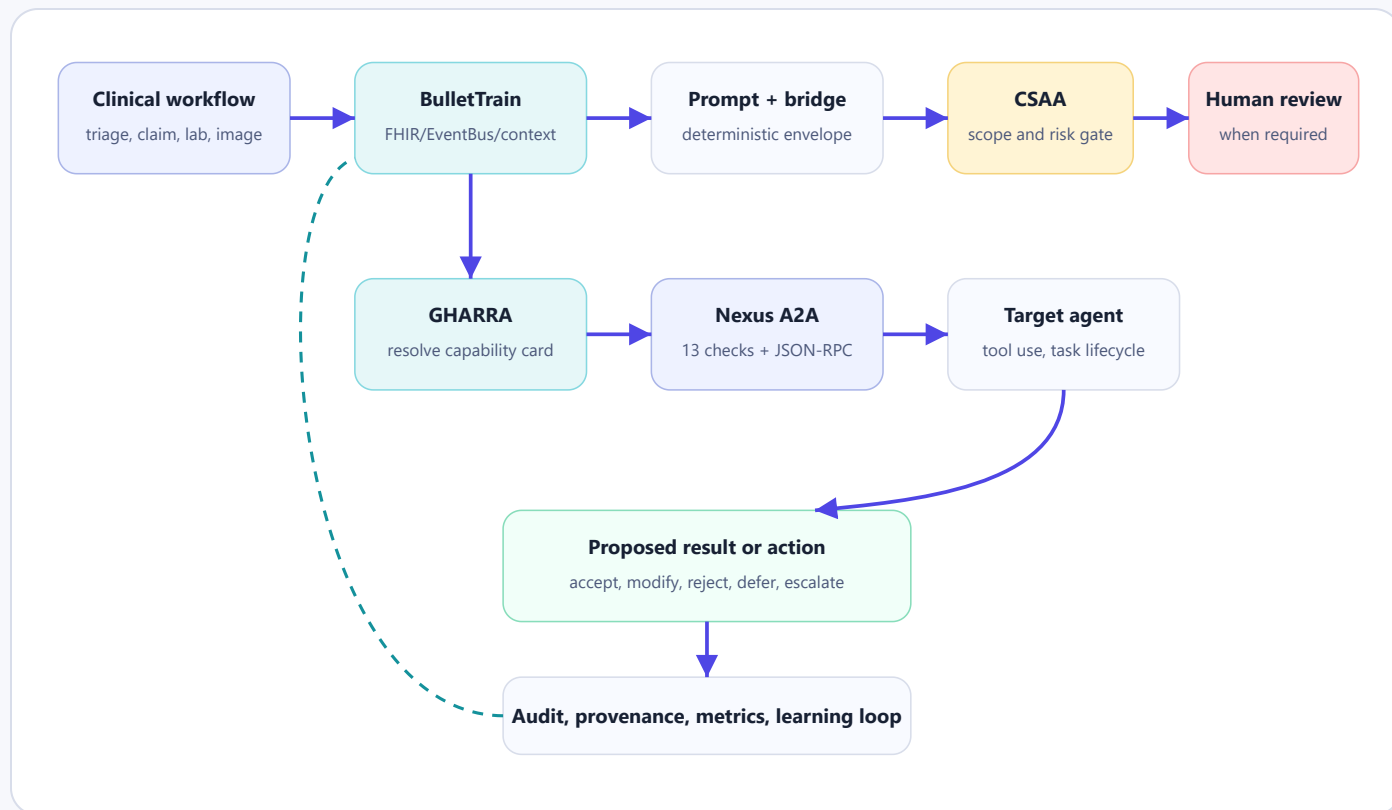
Runtime substrate

Private cloud, sovereign zones, regional cloud, Kubernetes, databases, queues, GPUs, CPU, storage, network

Boundary that matters. A clinical agent can recommend, draft, classify, route, or create a proposed update. It does not mutate sibling records directly unless the action is mediated by BulletTrain, authorised by policy, and auditable against the system of record.

Transport that matters. Nexus receives admitted GHARRA records and transports JSON-RPC calls. It is not the registry and it is not the business orchestrator.

How a clinical agent call should flow



Read the loop as a control contract. BulletTrain assembles trusted clinical context and records outcomes. Prompt Engine and Bridge SDK build deterministic, inspectable agent envelopes. CSAA decides the safety posture. GHARRA returns the eligible agent and trust bundle. Nexus opens transport only after admission checks. Human review is mandatory where the clinical or regulatory rule says so.

Repo-to-layer map with direct evidence

REPO OR FAMILY	MAPPED LAYER	REPO-BACKED RESPONSIBILITY	EVIDENCE STATUS	PRIMARY REFS
BulletTrain	Interoperability, identity, audit, AI grounding	FHIR R4-native HIE with 160+ services, API gateway, EventBus, SignalBox control plane, RBAC, audit, consent governance, terminology, DICOM, CDA, X12, OpenHIE, observability.	REPO EVIDENCE	README.md:33,43,51,60,65,67-69,784-785
global-agent-registry	Agent identity, registry, trust, discovery	GHARRA provides federated zero-trust agent discovery, routing advice, trust verification, zero-patient-data guardrails, A2A agent cards, JWT/mTLS/DPoP auth, and MCP exposure.	REPO EVIDENCE	README.md:26,91-101,139-166,209,259
nexus-a2a-protocol	Agent transport and admission	Nexus implements trusted agent-to-agent communication over JSON-RPC 2.0, 25 clinical agents, on-demand gateway, 13 route checks, and GHARRA metadata injection. Discovery and registry ownership stay outside Nexus.	REPO EVIDENCE	README.md:26,71-73,144-166; docs/gharra_integration.md:228-231
symphonix-bridge-sdk	Canonical envelopes and adapter translation	Bridge SDK provides canonical pivot translation, deterministic envelope serialization, audit and routing consistency, patient identity alignment, and a no-direct-sibling-call rule via BulletTrain hub mediation.	REPO EVIDENCE	docs/REQUIREMENTS.md:14,35,37; docs/architecture/patient-identity-closed-loop.md:21-30,130,318
prompt-engine	Prompt specs, templates, policies, optimization	Prompt Engine loads structured specs, assembles deterministic prompts, handles healthcare templates, PHI redaction, policy injection, prompt testing, and dispatches optimizer calls through BulletTrain hub mediation.	REPO EVIDENCE	docs/REQUIREMENTS.md:7-34,344-363,479-500,616-689; src/prompt_engine/optimizer/workflow.py:3-11,146-231

Assurance, strategy, and domain layers

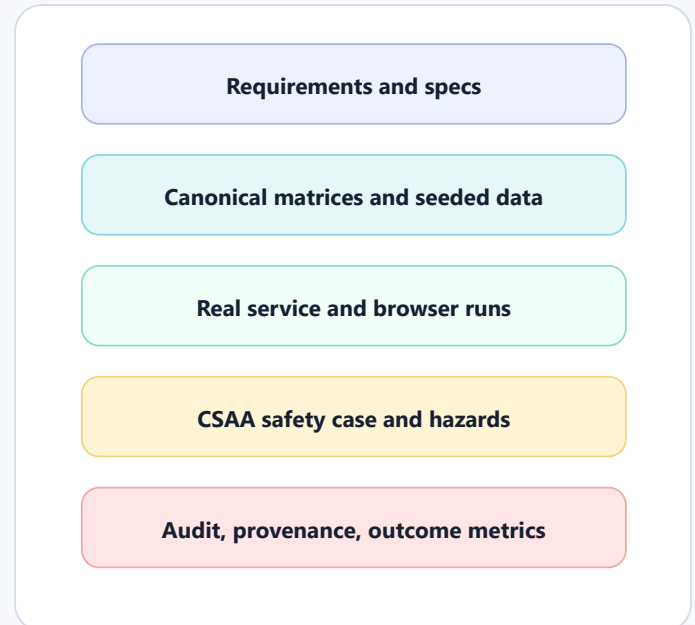
REPO OR FAMILY	MAPPED LAYER	REPO-BACKED RESPONSIBILITY	EVIDENCE STATUS	PRIMARY REFS
csaa	Clinical safety assurance	CSAA classifies safety scope, generates safety cases, hazard logs, clinical risk management plans, and release recommendations. It should gate clinical-agent production release.	REPO EVIDENCE	csaa/scope.py:1-16,142-217; csaa/case_builder.py:55-117,160-199
signalbox-mcp	Browser evidence, test execution, capture attestation	SignalBox provides MCP browser control, persona-aware tests, structured assertions, visual regression, scenario execution, screenshots, session recording, network and console inspection, and GHARRA-anchored capture manifests.	REPO EVIDENCE	versions/v2/README.md:8-20,41,103-145,181-227; INTEROP.md:9-15,101-110
caid-agent	SDLC orchestration, requirements, matrices, verification	CAID handles prompt-as-requirement traceability, IEEE documentation, canonical matrices, NFR derivation, multi-agent engineering orchestration, architecture advisory, memory, and evidence gates.	REPO EVIDENCE	README.md:31-56,92-127,235-247,725-728
health-agent-workspace	Market, pilot, and research grounding	Provides GHARRA healthcare AI market priorities, Horizon 1000 alignment, tiered build sequence, SE-agent literature review, and Kenya pilot requirements context.	STRATEGY EVIDENCE	GHARRA...md:8-10,360-381; literature_review...md:13-15,252-275
domain apps	Workflow action and system-of-record boundaries	Portals, triage, scheduling, LIS, PACS/RIS, pharmacy, EPS, e-claims, HMIS, and specialty systems own their bounded domain data and expose service contracts; cross-system movement is mediated by BulletTrain.	FAMILY EVIDENCE	agent-first.md:19,82,108-173; bullettrain-integration-doctrine.md:56,68,100,161

Architecture is not complete without evidence

Evidence stack

1. **Requirements.** Repo-level functional and non-functional specs, prompt-as-requirement trace, and seeded requirement ledgers define the target behaviour.
2. **Canonical matrices.** Functional scenarios use the 14-column BulletTrain matrix. CAID NFR scenarios use the 18-column matrix where required.
3. **Real-service validation.** Internal service evidence should use real seeded paths and real service ownership, not internal fakes.
4. **Browser and visual evidence.** SignalBox captures persona-aware browser runs, screenshots, assertions, and visual regressions.
5. **Clinical safety evidence.** CSAA provides scope classification, hazards, risk controls, safety case reports, and release decisions.
6. **Audit and provenance.** BulletTrain and Bridge SDK produce auditable envelopes, FHIR/AuditEvent/Provenance alignment, and replayable execution records.

Evidence diagram



Do not count stale substitutes as readiness. Internal fakes, mocks, stand-in backends, empty dashboards, synthetic counters, and scenario-only claims are not readiness evidence for an internal service. They can be useful during development, but the architecture report should mark them as gaps unless the seeded real path is verified.

Source-of-truth files for the architecture report

EVIDENCE CATEGORY	PRIMARY FILES	USE IN THIS REPORT
Strategy	symphonix-health-docs/strategy/agent-first.md symphonix-health-docs/strategy/bullettrain-integration-doctrine.md	Defines the additive agent stance, cross-system exchange rule, core platform repos, and clinical-agent governance gates.
Platform code/specs	BulletTrain/README.md global-agent-registry/README.md nexus-a2a-protocol/README.md nexus-a2a-protocol/docs/gharra_integration.md	Defines HIE fabric, registry, route admission, JSON-RPC transport, trust, and zero-PHI boundaries.
Assurance	csaa/csaa/scope.py csaa/csaa/case_builder.py signalbox-mcp/versions/v2/README.md symphonix-bridge-sdk/docs/REQUIREMENTS.md	Defines clinical safety scope, evidence capture, canonical envelopes, deterministic audit replay, and test evidence.

Recommended delivery slice

The report should not propose a full-platform build as the first step. The repo evidence points to a narrower first slice: one measurable workflow, BulletTrain-mediated context, CSAA safety classification, GHARRA registration, Nexus transport where agent-to-agent calls are needed, SignalBox evidence, and explicit human review.

PRIORITY	COMPONENT	ROLE IN FIRST SLICE	ACCEPTANCE EVIDENCE
P0	BulletTrain + domain API	Expose the workflow through a real service contract and BulletTrain-mediated context, not a direct agent-to-database path.	Real service test, EventBus/API evidence, audit event.
P0	prompt-engine + symphonix-bridge-sdk	Build deterministic prompts and canonical action envelopes with stable serialization, policy injection, PHI handling, and replayable audit.	Prompt regression, bridge requirements tests, envelope replay.
P0	csaa	Classify clinical scope and produce safety-case/hazard evidence before the workflow is described as production-ready.	Scope classification, hazard log, release decision.
P0	global-agent-registry	Register the agent capability card, trust material, zone metadata, PHI policy posture, and route policy.	GHARRA record, zero-PHI scan, trust bundle.
P1	nexus-a2a-protocol	Use Nexus only when the workflow requires agent-to-agent delegation. Validate the GHARRA record and transport JSON-RPC.	13-check admission result, JSON-RPC trace.
P1	signalbox-mcp	Capture browser, persona, visual, network, and assertion evidence for the workflow and attach it to the audit story.	Scenario result, screenshot, capture manifest.
P2	CAID + health-agent-workspace	Use CAID to keep requirements, matrices, NFRs, and documentation in sync; use Health Agent Workspace to choose market-aligned workflows and pilot constraints.	RTM, matrix coverage, SRS/pilot evidence.

Best first workflow candidates

- Prior authorization or insurance e-claims, because it is high volume and has clear human review points.
- Clinical pathway deviation or triage, because CSAA, HITL, provenance, and evidence capture are central.
- Scheduling or appointment automation, because it gives measurable cycle-time value with lower clinical risk.

Do not start here

- Do not start with a platform-wide "superpersona" that can mutate many systems.
- Do not route traditional clients through GHARRA/Nexus just to make them look agentic.
- Do not treat a diagram, scenario-only evidence, or generated readiness report as implementation proof.

Questions that remain genuinely unresolved

Deployment

- Which sovereign or regional hosting pattern is the first target: Kenya, UK, Ireland, Ghana, Rwanda, Nigeria, UAE, or a private lab environment?
- What are the first-slice latency, throughput, uptime, and offline-operation targets?
- Which identity provider and trust-anchor model will be used for the first tenant?

Clinical governance

- Which actions are advisory-only, which require human sign-off, and which are hard-stopped?
- What clinical safety scope does CSAA assign to the first workflow?
- Which audit records must be FHIR AuditEvent/Provenance versus internal operational audit?

Agent boundaries

- Which agent capabilities belong in GHARRA as signed cards, and which remain internal tools behind one capability?
- When is Nexus A2A mandatory versus unnecessary because the workflow only calls local service APIs?
- Which context data is safe to expose to a model, and which must stay in BulletTrain as references only?

Evidence

- Which SignalBox scenario proves the workflow from the correct persona?
- Which canonical matrix rows cover the same behaviour?
- Which real seeded service path replaces any previous internal fake or stand-in?

Design note. The revised diagrams intentionally keep GHARRA, Nexus, CSAA, BulletTrain, SignalBox, Prompt Engine, Bridge SDK, and CAID separate. Collapsing them into one "agent brain" would hide the controls that make the architecture reviewable.

Reference list for this revision

REFERENCE	WHY IT IS USED
<code>symphonix-health-docs/strategy/agent-first.md</code>	Defines additive agent-first posture, core repo roles, and clinical-agent governance gates.
<code>symphonix-health-docs/strategy/bullettrain-integration-doctrine.md</code>	Defines BulletTrain as the mandatory cross-system exchange, audit, consent, provenance, and AI grounding plane.
<code>BulletTrain/README.md</code>	Defines HIE scope, standards, 160+ microservices, SignalBox control plane, RBAC, audit, consent, and EventBus posture.
<code>global-agent-registry/README.md</code> and <code>CLAUDE.md</code>	Defines GHARRA as zero-trust, federated, zero-PHI registry and route-advisory layer.
<code>nexus-a2a-protocol/README.md</code> and <code>docs/gharra_integration.md</code>	Defines Nexus JSON-RPC transport, 25 agents, 13 route checks, and boundary with GHARRA and BulletTrain.
<code>symphonix-bridge-sdk/docs/REQUIREMENTS.md</code>	Defines canonical adapter translation, stable envelope serialization, and BulletTrain hub mediation constraints.
<code>prompt-engine/docs/REQUIREMENTS.md</code> and <code>src/prompt_engine/optimizer/workflow.py</code>	Defines prompt spec, policy, PHI, agentic mode, test, and BulletTrain hub dispatch behaviour.
<code>csaa/csaa/scope.py</code> and <code>csaa/csaa/case_builder.py</code>	Defines clinical scope classification, hazard/risk handling, safety case reports, and release recommendation logic.
<code>signalbox-mcp/versions/v2/README.md</code> and <code>INTEROP.md</code>	Defines browser-test tools, persona assertions, visual regression, scenario execution, and capture attestation.